Manoj Kumar Ashok

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Summary

Data Science professional with experience building machine learning models, statistical analysis workflows, and data pipelines using Python and SQL. Skilled in data preparation, feature engineering, model evaluation, visualization, and reproducible analysis. Experienced working across the machine learning lifecycle and translating complex data into clear, practical insights.

Technical Skills

Languages: Python, SQL, R
Machine Learning: Scikit-learn, PyTorch, XGBoost, TensorFlow
Data: Pandas, NumPy, PySpark, ETL, Apache Spark, Airflow
Databases & Cloud: PostgreSQL, MySQL, AWS (S3, Athena, EC2)
Tools: Git, Docker, Linux, Tableau, Power BI, Microsoft Excel, PowerPoint, Word

Experience

Research Assistant | DePaul University Jan 2025 – Mar 2026

- Developed and evaluated conditional GAN architectures, including IcGAN and RoCGAN, for image-generation research using Python and PyTorch.
- Built reproducible machine learning workflows for data ingestion, preprocessing, validation, model training, and experiment tracking.
- Cleaned and validated imaging and research datasets, identifying data-quality issues and automating repetitive quality-assurance checks.
- Evaluated model performance using precision, recall, F1-score, ROC-AUC, confusion matrices, FID, and Inception Score; prepared visualizations and technical summaries for faculty review.

Data Analyst | Zoho Corporation Feb 2023 – Dec 2023

- Analyzed operational and business data using SQL and Python to identify trends, investigate performance issues, and support recurring reporting requirements.
- Built and maintained 10 interactive dashboards and recurring reports in Zoho Analytics, tracking KPIs across business operations.
- Created SQL queries to extract, join, clean, and validate datasets containing over 100,000 records from multiple sources.
- Automated recurring data-preparation and reporting tasks using SQL and Python, reducing manual reporting effort by approximately 25%.
- Partnered with three internal teams to gather reporting requirements, define metrics, and translate business questions into dashboards and actionable insights.

Data Analyst Intern | Zoho Corporation Oct 2022 – Feb 2023

- Supported the preparation of business reports and dashboards using Zoho Analytics, SQL, and spreadsheet-based data sources.
- Wrote and modified SQL queries to retrieve, filter, aggregate, and validate data for ad hoc analysis and scheduled reporting.
- Cleaned and reconciled datasets containing over 25,000 records, resolving missing values, duplicates, and formatting inconsistencies.
- Assisted in creating five KPI dashboards and visual reports used by analysts and business stakeholders to monitor performance.

Projects

Local LLM Resume & Career Advisor | Python, LLaMA.cpp, CUDA, FastAPI, Streamlit, Docker *GitHub* 2025

- Built a local GPU-accelerated LLM application that analyzes resumes and generates career guidance without relying on external model APIs.
- Developed a retrieval-augmented generation pipeline using document chunking, embeddings, semantic retrieval, and structured prompt templates.
- Created FastAPI endpoints for document ingestion and inference and integrated them with a Streamlit user interface.
- Containerized the application with Docker and configured quantized LLaMA.cpp inference for reproducible local deployment.

Company Bankruptcy Risk Prediction | Python, SQL, Scikit-learn, XGBoost, PCA, SMOTE *GitHub* 2025

- Developed a bankruptcy-risk classification pipeline using 6,800+ company records and 96 financial indicators covering liquidity, profitability, leverage, and cash flow.
- Applied feature preprocessing, PCA, stratified sampling, class weighting, and SMOTE to address a highly imbalanced target variable.
- Compared multiple classification models and achieved 79.5% minority-class recall and 0.949 ROC-AUC with XGBoost.
- Used SQL and Python to validate source data, investigate missing values and outliers, and prepare reproducible model-evaluation outputs.

NYC Taxi Analytics & A/B Testing Simulation | Python, PySpark, SQL, AWS Athena, NumPy 2024–2025

- Processed and analyzed more than 3 million NYC taxi-trip records using PySpark, SQL, and AWS Athena.
- Performed data cleaning and feature engineering to examine trip patterns, fare behavior, and factors associated with passenger tipping.
- Built a simulation framework for A/B experiments and calculated statistical significance, lift, conversion rate, and retention metrics.
- Automated data-quality checks and statistical reporting using reusable Python and SQL workflows.